

CHAPTER 1: CLOUD COMPUTING



What is Cloud Computing?

Cloud Computing means **using the internet to store, manage, and process data**, instead of using your own computer or local servers.

In simple words —

It's like renting a computer or storage space on the internet instead of buying your own.



Example

- When you use **Google Drive** to store files or **Netflix** to watch movies online — you are using **cloud computing**.
- The data and programs are stored on **remote servers (the cloud)**, not on your device.



How Cloud Computing Works

1. You connect to the internet.
2. You use a **cloud service provider** (like AWS, Google Cloud, or Microsoft Azure).
3. The provider gives you access to computing power, storage, or applications online.
4. You can use it from **anywhere, anytime**.



Types of Cloud Computing Services

1. IaaS (Infrastructure as a Service)

- Provides **virtual hardware** like servers and storage.
- Example: Amazon Web Services (AWS), Google Cloud Platform.

2. PaaS (Platform as a Service)

- Provides tools for **developing and testing software**.
- Example: Microsoft Azure, Heroku.

3. SaaS (Software as a Service)

- Provides **ready-to-use software** over the internet.
- Example: Gmail, Zoom, Google Docs.



Types of Cloud Deployment

1. Public Cloud

- Services shared by many users.
- Example: Google Drive.

2. Private Cloud

- Used by one organization only.
- Example: Company's internal data servers.

3. Hybrid Cloud

- Combination of public and private clouds.
- Used for both flexibility and security.



Advantages of Cloud Computing

- 💰 **Cost-effective** – No need to buy expensive hardware.
- 🌐 **Accessible anywhere** – Use from any device with internet.
- ⚡ **Scalable** – Increase or decrease resources easily.
- 🛡️ **Secure** – Data backup and disaster recovery are built in.
- 🔄 **Automatic Updates** – Providers keep everything up to date.



Disadvantages of Cloud Computing

- 🌐 **Needs internet** – No access without connection.
- 🛒 **Data security concerns** – Your data is stored on someone else's servers.
- ⚙️ **Limited control** – The service provider manages the infrastructure.



In Short

Cloud Computing = Internet-based computing

It lets you **store, access, and run applications online** without needing physical devices or storage.



Need of Cloud Computing

Cloud computing is needed because it makes using technology **easier, faster, cheaper, and more flexible**.

Instead of buying and maintaining your own computers or servers, you can use powerful online resources whenever you need them.



1. Cost Savings

- No need to buy expensive hardware or software.
- You only **pay for what you use**.
- Reduces electricity and maintenance costs.

2. Speed and Performance

- Cloud services can be set up in **minutes**.
- You get **high-speed performance** because providers use powerful servers.
- Software updates and upgrades happen automatically.

3. Accessibility and Mobility

- You can **access your data and applications anytime, anywhere** using the internet.
- Helps people work from home, school, or office easily.
- Useful for **remote work and online learning**.

4. Scalability and Flexibility

- You can **increase or decrease** resources (like storage or computing power) as needed.
- Very useful for growing businesses or seasonal workloads.

5. Data Security and Backup

- Cloud providers offer **data backup, recovery, and encryption**.
- Even if your computer crashes, your data stays safe in the cloud.
- Protects against data loss and disasters.

6. Focus on Core Business

- Companies don't need to worry about managing IT infrastructure.
- They can focus on **innovation, growth, and services** instead.

7. Collaboration and Sharing

- Cloud allows multiple users to **work on the same file or project at the same time**.
- Example: Google Docs lets many people edit a document together.

8. Automatic Updates

- Cloud service providers automatically **update software and security systems**.
- Users always have the latest versions without extra effort.

In Short

We use **cloud computing** because it is **cost-effective, fast, flexible, and secure**. It helps individuals and organizations **store data, access services, and collaborate easily** through the internet.

Characteristics of Cloud Computing

Cloud computing has some important features that make it powerful, flexible, and easy to use. These characteristics explain **why cloud computing is so popular** in today's world.

1. On-Demand Self-Service

- Users can use cloud resources like storage, software, or servers **whenever they need** — without asking anyone.
- Example: You can create a new Gmail account or upload files to Google Drive instantly.

2. Broad Network Access

- Cloud services are available **through the internet** and can be accessed from **any device** — computer, phone, or tablet.
- You just need an internet connection.

3. Resource Pooling

- Cloud providers use shared servers to serve multiple customers.
- Resources like storage, memory, and processing power are **pooled and shared efficiently** among users.
- Each user's data stays **separate and secure**.

4. Rapid Elasticity (Scalability)

- Cloud resources can be **increased or decreased** easily depending on your needs.
- Example: A website can handle more visitors during sales or festivals by quickly adding more servers.

5. Measured Service (Pay-as-You-Go)

- You **pay only for what you use**, just like electricity or water.
- The system automatically tracks how much computing power, storage, or bandwidth you use.

6. Automatic Updates and Maintenance

- Cloud providers automatically **update software, fix bugs, and add new features**.
- Users don't have to worry about installation or maintenance.

7. Security and Reliability

- Cloud systems use **data encryption, backups, and multiple servers** to keep information safe and always available.
- Even if one server fails, data remains secure and accessible.

8. Multi-Tenancy

- Many users (tenants) can share the same cloud infrastructure while keeping their data separate.
- This helps in **reducing costs and improving efficiency**.

In Short

Cloud computing is **on-demand, flexible, and cost-effective**.

Its main characteristics are **self-service, accessibility, scalability, resource sharing, and reliability**.

Disadvantages of Cloud Computing

Even though cloud computing has many benefits, it also has some **drawbacks and risks** that users should know about.

1. Requires Internet Connection

- Cloud services work **only with an active internet connection**.
- If the internet is slow or unavailable, you **cannot access your data or applications**.
- Example: You can't open Google Drive files without internet.

2. Data Security and Privacy Issues

- Your data is stored on **someone else's servers** (the provider's).
- There is always a **risk of hacking, data leaks, or unauthorized access**.
- Sensitive information must be protected carefully.

3. Limited Control

- Users have **less control** over the cloud infrastructure and settings.
- The provider manages hardware, updates, and security — not the user.
- You depend on the provider's policies.

4. Ongoing Costs

- While cloud services reduce hardware costs, they often involve **monthly or yearly fees**.
- Over time, this can become **more expensive** for large-scale users.



5. Downtime and Service Outages

- Sometimes cloud servers may **go down** due to technical problems or maintenance.
- This can **interrupt access** to data or applications temporarily.



6. Data Transfer and Bandwidth Limits

- Uploading or downloading large files can be **slow and costly** if your internet has limited bandwidth.
- Some providers charge extra for **high data usage**.



7. Vendor Lock-In

- Moving data from one cloud provider to another can be **difficult and time-consuming**.
- Each provider may use **different formats and systems**.



In Short

Cloud computing is convenient and powerful, but it depends on the internet and the provider's security.

The main disadvantages are **internet dependency, security risks, limited control, and possible high costs**.



Cloud Computing Service Providers

Cloud computing service providers are companies that **offer cloud-based services** such as storage, servers, software, and databases over the internet.

They allow users and businesses to **store data, run applications, and manage IT resources** without owning physical hardware.



1. Amazon Web Services (AWS)

- **Developed by:** Amazon
- **Type:** Public Cloud
- **Services:** Storage (S3), Computing (EC2), Databases, AI tools, etc.
- **Features:** Reliable, secure, and used by millions of companies worldwide.
- **Example:** Netflix and LinkedIn use AWS.



2. Microsoft Azure

- **Developed by:** Microsoft

- **Services:** Virtual machines, databases, app hosting, and AI tools.
- **Features:** Works well with Windows and Microsoft products like Office 365.
- **Example:** Used by companies like BMW and Samsung.

3. Google Cloud Platform (GCP)

- **Developed by:** Google
- **Services:** Cloud storage, machine learning, data analytics, and app development.
- **Features:** Known for speed, security, and AI capabilities.
- **Example:** Used by Spotify and Snapchat.

4. IBM Cloud

- **Developed by:** IBM
- **Services:** Cloud computing, AI (Watson), and blockchain technology.
- **Features:** Focuses on enterprise (large business) solutions and hybrid clouds.

5. Oracle Cloud

- **Developed by:** Oracle Corporation
- **Services:** Databases, software applications, and infrastructure.
- **Features:** Best known for its strong **database management services**.

6. Alibaba Cloud

- **Developed by:** Alibaba Group (China)
- **Services:** Storage, computing, big data, and cybersecurity.
- **Features:** Leading cloud provider in Asia.

7. VMware Cloud

- **Developed by:** VMware
- **Services:** Virtualization and cloud infrastructure solutions.
- **Features:** Helps businesses move existing systems to the cloud easily.

In Short

Cloud service providers like **AWS, Azure, and Google Cloud** offer storage, software, and computing power through the internet. They help users **save money, scale quickly, and access technology from anywhere**.



Evolution of Cloud Computing

Cloud computing didn't appear overnight — it developed **step by step over many years** as technology improved.

Let's see how it evolved from the early days of computing to today's modern cloud systems.



1. Early Stage (1950s – 1960s): Mainframe Computing

- Large computers called **mainframes** were used by big companies.
- Many users shared one powerful machine using “**time-sharing**” systems.
- This idea of **sharing computing power** was the **foundation of cloud computing**.



2. Virtualization Era (1970s – 1990s)

- **Virtual machines (VMs)** were introduced — allowing one physical computer to act like many.
- Companies like IBM developed systems to **divide hardware resources efficiently**.
- This made computing more **flexible and cost-effective**.



3. Internet and Web Growth (1990s)

- The **Internet became popular**, connecting millions of computers worldwide.
- Companies started offering **online services and storage**.
- Example: Email services like Yahoo! Mail and Hotmail used early forms of the cloud.



4. Birth of Cloud Computing (2000s)

- The term “**Cloud Computing**” became popular around **2006**.
- **Amazon launched AWS (Amazon Web Services)**, offering storage and computing power online.
- Other big players like **Google, IBM, and Microsoft** joined soon after.
- This was the beginning of **modern cloud services**.



5. Expansion and Modern Cloud (2010s – Present)

- Cloud computing grew rapidly and became **essential for businesses and individuals**.
- New services appeared:
 - **SaaS** – Software as a Service (e.g., Gmail, Zoom)
 - **PaaS** – Platform as a Service (e.g., Heroku, Google App Engine)
 - **IaaS** – Infrastructure as a Service (e.g., AWS, Azure)
- **Hybrid and multi-cloud systems** were introduced for better flexibility.



6. Future of Cloud Computing (Now and Beyond)

- Cloud is becoming smarter with **AI, machine learning, and edge computing**.
- More focus on **security, automation, and sustainability**.
- Used in **IoT (Internet of Things)**, healthcare, education, and more.

In Short

Cloud computing evolved from **mainframe computers** to **virtualization**, then to **internet-based services**, and finally to **advanced global cloud platforms** we use today. It continues to grow with **AI, automation, and global connectivity**.

Flow Chart: Evolution of Cloud Computing

1950s–1960s



 Mainframe Computing

- Time-sharing systems
- Many users shared one main computer



1970s–1990s

 Virtualization Era

- Introduction of virtual machines (VMs)
- One computer acted like many
- Improved flexibility and cost efficiency



1990s

 Internet and Web Growth

- Rise of the Internet
- Online services like email and storage
- Early form of cloud services



2000s

 Birth of Cloud Computing

- Term “Cloud Computing” introduced
- Amazon Web Services (AWS) launched (2006)
- Google, IBM, Microsoft entered the field



2010s–Present

 Modern Cloud Computing

- SaaS, PaaS, IaaS services introduced
- Hybrid and multi-cloud systems

- Used in business, education, healthcare



Future

 Intelligent Cloud

- AI and Machine Learning integration
- Edge computing and IoT
- Focus on security and automation

 **In Short**

Cloud computing evolved from **mainframes** → **virtualization** → **internet** → **cloud services** → **intelligent cloud**.

Evolution of Cloud Computing



Mainframe Computing

- Time-sharing systems
- Many users shared one main computer



Virtualization Era

- Introduction of virtual machines (VMs)
- One computer acted like many



Internet and Web Growth

- Rise of the Internet
- Online services like email and storage



Birth of Cloud Computing

- Term "Cloud Computing"
- Amazon Web Services (AWS) launched (2006)



Modern Cloud Computing

- SaaS, PaaS, IaaS services introduced
- Hybrid and multi-cloud systems



Intelligent Cloud

- AI and Machine Learning integration
- Edge computing and IoT
- Focus on security and automation

Cloud Computing Architecture

Cloud Computing Architecture is the **structure or design** of all the components that make cloud services work.

It includes **front-end (what users see)** and **back-end (what happens behind the scenes)** parts that work together through the **internet**.

1. Components of Cloud Computing Architecture

Cloud computing architecture mainly has **two parts**:

A. Front-End

- It is the **user side** of cloud computing.
- This is what users interact with when they use cloud services.
- Example: The interface of Google Drive, Dropbox, or Gmail.

Main Elements:

- Web browser or application (used to access cloud)
- User interface (UI)
- Client devices (computers, mobiles, tablets)

B. Back-End

- It is the **cloud provider's side** that stores data, runs programs, and manages services.
- This part is hidden from users.

Main Elements:

1. **Servers** – Handle requests and process data.
2. **Storage** – Stores files, databases, and backups.
3. **Database Management** – Organizes and retrieves information efficiently.
4. **Application** – Software that provides services (like Gmail, Zoom, or AWS EC2).
5. **Security and Monitoring** – Protects data and ensures smooth performance.

2. Cloud Delivery Models

These are the **types of services** provided through cloud computing:

IaaS (Infrastructure as a Service)

- Provides virtual hardware like servers and storage.
- Example: AWS EC2, Google Compute Engine.

PaaS (Platform as a Service)

- Provides a platform to **build and run applications**.

- Example: Microsoft Azure, Heroku.



SaaS (Software as a Service)

- Provides **ready-to-use software** over the internet.
- Example: Gmail, Google Docs, Zoom.



3. Cloud Deployment Models

These show **how the cloud is set up and used**:

1. **Public Cloud** – Shared by many users (e.g., Google Drive).
2. **Private Cloud** – Used by one organization only.
3. **Hybrid Cloud** – Mix of public and private clouds.
4. **Community Cloud** – Shared by organizations with similar needs.



4. How It Works (Simplified)

1. The user sends a request (like uploading a file).
2. The internet connects the user (front-end) to the provider (back-end).
3. The provider processes the request using servers and databases.
4. The result (like a stored file or app output) is sent back to the user.



5. Key Features

- **Scalability** – Add or reduce resources easily.
- **Security** – Data encryption and monitoring.
- **Automation** – Self-managed updates and maintenance.
- **Reliability** – Backup and recovery systems.



In Short

Cloud Computing Architecture is made of **front-end (user side)** and **back-end (provider side)**, connected through the **internet**, delivering services like **IaaS, PaaS, and SaaS** to users securely and efficiently.

CLOUD COMPUTING ARCHITECTURE

1. COMPONENTS OF CLOUD COMPUTING ARCHITECTURE

A. FRONT-END



User Interface (UI)

Client devices

B. BACK-END



Servers



Storage



Database Management



Application



Security and Monitoring

2. CLOUD DELIVERY MODELS



IaaS

Infrastructure as a Service

AWS EC2



PaaS

Platform as a Service

Microsoft Azure



SaaS

Software as a Service

Gmail

3. CLOUD DEPLOYMENT MODELS



Public Cloud



Private Cloud



Hybrid Cloud



Community Cloud

4. HOW IT WORKS (SIMPLIFIED)

1

User sends a request



Internet connects user to the provider



Server processes the request



Result is sent back to user

Applications of Cloud Computing

Cloud computing is used in **many areas of our daily life and business** because it provides storage, computing power, and software through the internet. Let's look at the main applications where cloud computing is most useful.

1. Data Storage and Backup

- Cloud helps store a large amount of data **securely online**.
- You can access files **anytime, anywhere**.
- It also provides **automatic backups**, preventing data loss.
- **Examples:** Google Drive, Dropbox, OneDrive.

2. Software as a Service (SaaS)

- You can use **ready-made software** over the internet without installing it.
- Saves time, money, and maintenance.
- **Examples:** Gmail, Zoom, Microsoft 365, Google Docs.

3. Business Applications

- Companies use cloud platforms for **managing data, customers, and finances**.
- Helps in **remote work, project management, and real-time collaboration**.
- **Examples:** Salesforce, Slack, Trello.

4. Education and E-Learning

- Students and teachers use cloud tools for **online classes, assignments, and file sharing**.
- Enables **remote learning** and **collaborative projects**.
- **Examples:** Google Classroom, Microsoft Teams, Zoom.

5. Big Data Analysis

- Cloud provides powerful tools for **analyzing large amounts of data**.
- Businesses use it to understand trends, customer behavior, and make decisions.
- **Examples:** AWS Big Data, Google BigQuery.

6. Entertainment and Media

- Used in **streaming movies, music, and games** directly from the cloud.
- No need to download large files — everything runs online.

- **Examples:** Netflix, Spotify, YouTube, PlayStation Cloud Gaming.



7. Healthcare

- Doctors and hospitals use cloud systems to **store patient records** and access them safely.
- Enables **telemedicine**, remote diagnosis, and faster treatment.
- **Examples:** Health Cloud, IBM Watson Health.



8. E-Commerce

- Online shopping websites use the cloud to **manage customer data, payments, and inventory**.
- Ensures **smooth performance** even during heavy traffic.
- **Examples:** Amazon, Flipkart, eBay.



9. Development and Testing

- Developers use cloud platforms to **build, test, and deploy applications** quickly.
- No need for expensive hardware.
- **Examples:** Microsoft Azure, Google App Engine, AWS Lambda.



10. Government and Public Services

- Governments use cloud systems for **citizen data management, online services, and digital records**.
- Improves **efficiency and transparency**.



In Short

Cloud computing is used in **education, business, healthcare, entertainment, data storage, and app development**.

It makes work **faster, cheaper, and accessible from anywhere** in the world.



What is Parallel Computing?

Parallel Computing is a type of computing where **many tasks or processes are carried out at the same time** (simultaneously) to solve a problem faster.

In simple words —

Instead of one computer or processor doing all the work alone, **many processors work together** to complete the task more quickly.



Example

Imagine you have to clean a big room:

- If **one person** does it, it will take a long time.
- If **four people** clean different parts **at the same time**, the work finishes faster.

👉 That's how **parallel computing** works!



Need for Parallel Computing

- To **save time** and complete large tasks faster.
- To **handle big data** and complex problems.
- To **increase efficiency** of computers and servers.



Steps in Parallel Computing

Parallel computing usually follows **five main steps** 📌



1. Problem Decomposition

- The main problem is **divided into smaller sub-problems**.
- Each sub-problem can be solved separately.
- Example: Dividing a big file into smaller parts for processing.



2. Task Distribution

- The smaller tasks are **assigned to different processors or computers**.
- Each processor gets its own piece of work to do.



3. Parallel Execution

- All processors **work at the same time** on their assigned tasks.
- This is the main stage where the actual computation happens simultaneously.



4. Synchronization

- After all processors complete their tasks, the system **synchronizes (waits)** until all are done.
- Ensures that all partial results are ready before combining them.



5. Result Combination

- The results from all processors are **merged together** to form the **final output**.
- Example: Combining the smaller processed parts into a single completed file.



Types of Parallel Computing



1. Data Parallelism

- Same task is performed on **different pieces of data** at once.

◆ 2. Task Parallelism

- **Different tasks** are performed **at the same time** by different processors.

◆ 3. Hybrid Parallelism

- Combination of both data and task parallelism.



Advantages of Parallel Computing

- ✓ Faster execution time
- ✓ Better performance for large problems
- ✓ Efficient use of hardware resources
- ✓ Saves time in data-heavy tasks like AI, simulations, and big data analysis



Disadvantages

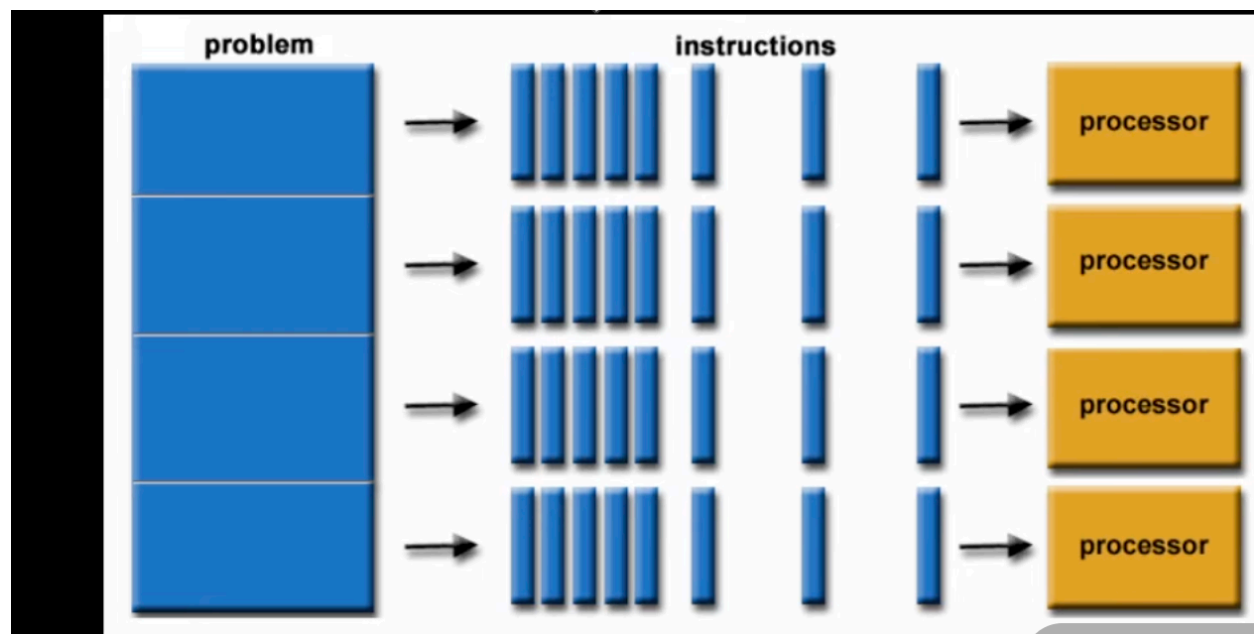
- ✗ Complex to design and manage
- ✗ Expensive hardware required
- ✗ Difficult to debug and synchronize



In Short

Parallel Computing means dividing a big problem into smaller parts and solving them **at the same time** using multiple processors.

Steps: **Decomposition** → **Distribution** → **Execution** → **Synchronization** → **Combination**





Types of Parallelism in Parallel Computing

In **Parallel Computing**, different parts of a program or data can be processed **at the same time**.

The way the work is divided is called **parallelism**.

There are mainly **two main types of parallelism** — **Data Parallelism** and **Task Parallelism** (and sometimes a mix of both).



1. Data Parallelism



Meaning:

- In **data parallelism**, the **same task** (operation) is performed **on different pieces of data** at the same time.
- Each processor works on a **separate chunk of data**, but all follow the **same instructions**.



Example:

- Suppose you want to add 10 numbers.
 - Processor 1 adds numbers 1–5
 - Processor 2 adds numbers 6–10
 - Both work **together** to finish faster.



Used In:

- Image processing
- Machine learning
- Scientific simulations



Advantages:

- Easy to implement
- Increases speed for large data sets



Disadvantage:

- Works best only when the **same task** is applied to all data.



2. Task Parallelism



Meaning:

- In **task parallelism**, **different tasks or functions** are executed **at the same time** on **different processors**.
- Each processor may be doing a **different job**.



Example:

- In a computer game:
 - One processor handles sound 🎵
 - Another controls graphics 🎮
 - Another manages user input ⌨️
 - All run **in parallel** to make the game smooth.



Used In:

- Real-time systems
- Complex applications (like AI or robotics)
- Web servers and multitasking programs



Advantages:

- Makes full use of multiple processors
- Suitable for systems doing different types of work



Disadvantage:

- Harder to manage and coordinate different tasks



3. Hybrid Parallelism



Meaning:

- **Hybrid Parallelism** is a **combination** of **data parallelism** and **task parallelism**.
- Some processors handle **different tasks**, and within each task, data is **processed in parallel**.



Example:

- In a weather simulation:
 - One group of processors predicts **temperature**, another predicts **rainfall**.
 - Inside each group, multiple processors handle **different regions** of data.



Used In:

- Supercomputers
- Scientific and engineering research
- Big data analytics



In Short

Type	What It Does	Example
Data Parallelism	Same task on different data	Adding numbers in parts
Task Parallelism	Different tasks at the same time	Game: sound, graphics, input
Hybrid Parallelism	Mix of both	Weather forecasting systems

Summary:

- **Data Parallelism:** Same work, different data.
- **Task Parallelism:** Different work, same time.
- **Hybrid Parallelism:** Combination of both for better performance.



What is Distributed Computing?

Distributed Computing is a type of computing where **many computers work together** to solve a large problem.

Each computer (called a **node** or **system**) works on a **part of the task**, and all results are **combined** to get the final output.

In simple words —

It means **dividing work among multiple computers** that are connected through a **network** (like the internet or LAN).



Example

Imagine you have to solve 1,000 math problems:

- One computer would take a long time.
- But if you divide them among **10 computers**, each does 100 problems — and all finish faster!

👉 That's **distributed computing**.



Main Features of Distributed Computing

◆ 1. Multiple Systems

- It uses **many interconnected computers** (nodes).
- Each computer works **independently** but cooperates with others.

◆ 2. Resource Sharing

- All computers share resources like data, files, and processing power.

◆ 3. Communication

- Systems **communicate through a network** (such as the internet).

◆ 4. Scalability

- More computers can be added easily to increase performance.

◆ 5. Fault Tolerance

- If one computer fails, others can continue working — system doesn't stop completely.



Components of Distributed Computing

1. **Clients** → Request services or resources.
2. **Servers** → Provide the requested services.
3. **Network** → Connects all systems together.
4. **Database** → Stores data shared among systems.



Steps in Distributed Computing

1. **Problem Division** – Break the main task into smaller parts.
2. **Distribution** – Send each part to different computers.
3. **Processing** – Each computer performs its assigned task.
4. **Coordination** – Systems communicate and share progress.
5. **Integration** – Combine all results to form the final output.



Advantages of Distributed Computing

-  **Faster processing** — many computers work at once.
-  **Cost-effective** — uses existing computers instead of one powerful machine.
-  **Reliable** — system continues even if one node fails.
-  **Easily expandable** — add more computers when needed.



Disadvantages of Distributed Computing

-  **Complex management** — harder to coordinate multiple systems.
-  **Security risks** — data travels over a network.
-  **Communication delays** — network speed can affect performance.
-  **Software compatibility issues** between systems.



Examples of Distributed Computing Systems

-  **Google Search Engine** – uses thousands of computers to index web pages.
-  **SETI@Home** – used home computers to analyze space signals.
-  **Cloud Computing Systems** – like AWS, Azure, and Google Cloud.
-  **Online Multiplayer Games** – use distributed servers to handle users worldwide.



In Short

Distributed Computing means using **multiple computers connected by a network** to work on **different parts of a task simultaneously**, combining their results to get the final answer.



Difference Between Parallel Computing and Distributed Computing

Both **Parallel Computing** and **Distributed Computing** aim to perform tasks faster by using multiple processors or systems.

But they differ in **how** and **where** the work is done.



1. Basic Definition

- **Parallel Computing:**
Multiple processors work **together on the same computer** to solve a problem **simultaneously**.
- **Distributed Computing:**
Multiple **independent computers** work **over a network** to solve parts of a problem.



2. System Type

- **Parallel Computing:**
Uses **one single system** with many processors (shared memory).
- **Distributed Computing:**
Uses **many separate systems (computers)** connected through a network.



3. Communication

- **Parallel Computing:**
Processors communicate **through shared memory** (fast communication).
- **Distributed Computing:**
Systems communicate **through a network** (like the internet or LAN).



4. Resource Sharing

- **Parallel Computing:**
All processors share **common resources** like memory and disk.
- **Distributed Computing:**
Each system has its **own memory and resources**.



5. Speed

- **Parallel Computing:**
Very **fast**, as communication happens within one system.
- **Distributed Computing:**
Slightly **slower**, because communication happens over a network.

6. Fault Tolerance

- **Parallel Computing:**
If one processor fails, the whole system may **stop working**.
- **Distributed Computing:**
If one system fails, **others keep working** — more reliable.



7. Scalability

- **Parallel Computing:**
Limited scalability — can add only a few processors to one system.
- **Distributed Computing:**
Highly scalable — can add **many computers** to the network easily.



8. Cost

- **Parallel Computing:**
Expensive, as it needs special high-performance hardware.
- **Distributed Computing:**
Cost-effective, as it uses normal computers connected together.



9. Example

- **Parallel Computing:**
Supercomputers, weather prediction models, scientific simulations.
- **Distributed Computing:**
Google Search Engine, online banking, cloud computing systems.



10. Data Handling

- **Parallel Computing:**
Handles **a single large problem** divided among processors.
- **Distributed Computing:**
Handles **many smaller tasks** that may be different but related.

In Short (Summary Table)

Point	Parallel Computing	Distributed Computing
1. Definition	Many processors in one system	Many systems over a network
2. System Type	Single computer	Multiple computers
3. Communication	Shared memory	Network-based
4. Resources	Shared	Independent
5. Speed	Faster	Slower

6. Fault Tolerance	Low	High
7. Scalability	Limited	High
8. Cost	Expensive	Cheaper
9. Example	Supercomputer	Google Search
10. Data Handling	One big task	Many small tasks

✓ In Simple Words

- **Parallel Computing** = Many processors inside **one machine** working together.
- **Distributed Computing** = Many **machines** connected through a **network**, sharing work.

☁ What is Cloud Elasticity?

Cloud Elasticity means the **ability of a cloud system to automatically increase or decrease its resources** (like storage, memory, or processing power) based on the user's needs.

In simple words —

Cloud Elasticity allows a system to **grow when demand increases** and **shrink when demand decreases**, automatically.

💡 Example

Imagine you run a shopping website:

- During a big sale, thousands of people visit at once — your system needs **more servers**.
- After the sale, visitors drop — now you can **reduce servers** to save cost.

👉 The cloud does this **automatically** using **elasticity**.

⚙ How Cloud Elasticity Works

1. **Monitors usage:** The cloud continuously checks how much computing power is needed.
2. **Adjusts resources:** It automatically adds or removes servers, storage, or bandwidth.
3. **Balances performance and cost:** You get the right amount of resources at the right time — no waste.

🧩 Types of Cloud Elasticity

◆ 1. Horizontal Elasticity (Scaling Out/In)

- Adds or removes **multiple instances** (servers or virtual machines).
- Example: Adding more servers during peak load.

◆ 2. Vertical Elasticity (Scaling Up/Down)

- Increases or decreases the **power** (CPU, RAM) of an existing machine.
- Example: Upgrading one server's memory during high demand.



Advantages of Cloud Elasticity

- **Cost Efficiency:** Pay only for what you use — no wasted money.
- **Better Performance:** Always have enough resources for current demand.
- **Automation:** No manual setup — the system adjusts itself.
- **Business Continuity:** Prevents crashes or slowdowns during high traffic.
- **Flexibility:** Easy to handle sudden changes in workload.



Disadvantages

- Requires advanced monitoring and management tools.
- May cause **temporary delays** during scaling.
- Not all applications can adapt to changing resources easily.



In Short

Cloud Elasticity = The ability of the cloud to **automatically scale resources up or down** based on demand — ensuring **efficiency, cost savings, and smooth performance**.



Example Services Offering Elasticity

- Amazon Web Services (AWS Auto Scaling)
- Microsoft Azure Scale Sets
- Google Cloud Autoscaler



What is Cloud Scalability?

Cloud Scalability means the **ability of a cloud system to grow or shrink its resources (like storage, servers, or processing power)** according to business needs — **manually or automatically**.

In simple words —

Cloud Scalability allows an organization to **increase or decrease computing resources** as demand changes, **without affecting performance**.



Example

- A startup begins with a few users → needs **small storage and servers**.
- As the business grows → cloud automatically **adds more resources** to handle new users.
- During slow periods → it can **reduce resources** to save money.

👉 This flexibility is called **Scalability**.



Why Cloud Scalability is Important

- 🌐 Helps handle **increasing users or workload** smoothly.
- 💰 **Saves cost** — pay only for what you use.
- ⚡ **Improves performance** — system stays fast even under high load.
- 🧠 **Supports business growth** without changing infrastructure.



Types of Cloud Scalability

Cloud scalability can be divided into **three main types** 📌



1. Vertical Scalability (Scaling Up/Down)

Meaning:

- Adding or removing **resources within a single server**.
- You make one machine more powerful by increasing CPU, RAM, or storage.

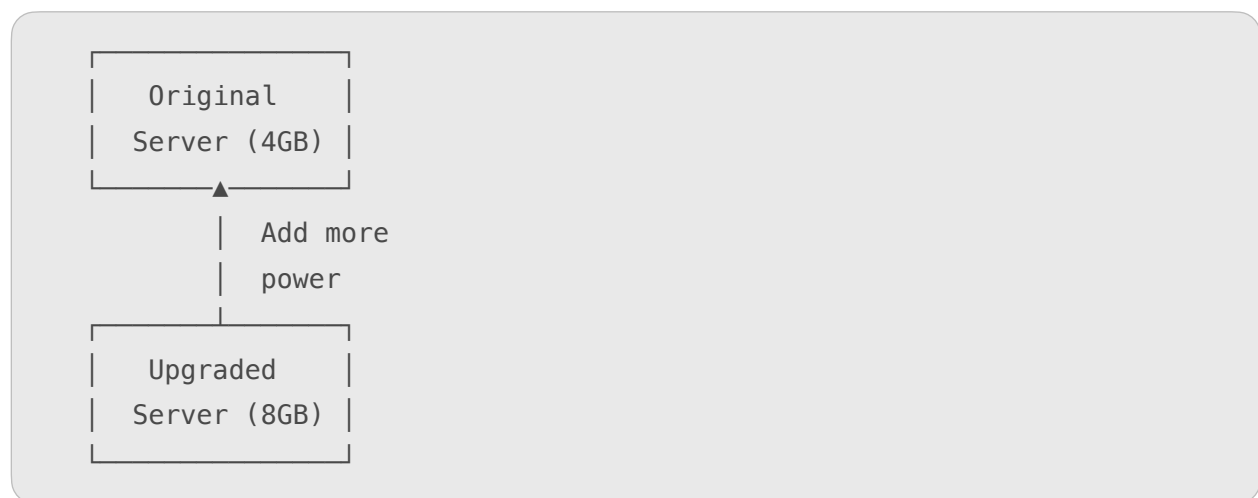
Example:

- Upgrading a 4GB RAM server to 8GB RAM for better performance.

Used In:

- Databases and applications that need high performance from one server.

Diagram: 📌



2. Horizontal Scalability (Scaling Out/In)

Meaning:

- Adding or removing **multiple servers** to handle more load.
- Instead of upgrading one machine, you **add more machines**.

Example:

- Adding more web servers to handle more visitors on a website.

Used In:

- Cloud environments like AWS, Google Cloud, and Azure.

Diagram: 📌

◆ 3. Diagonal Scalability (Combination of Both)

Meaning:

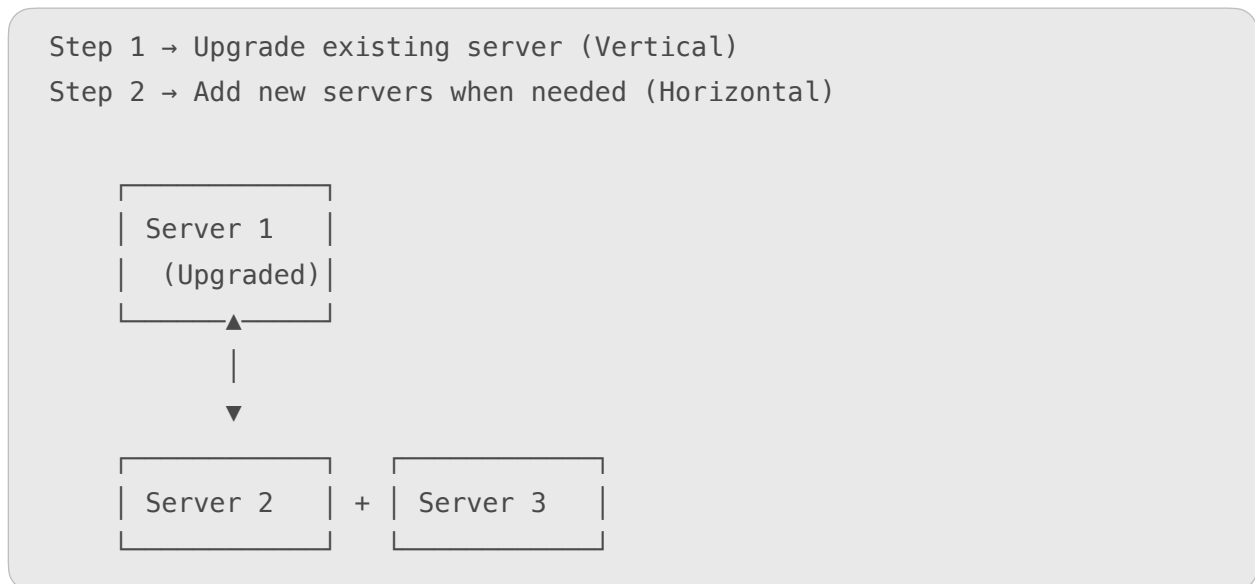
- A mix of **vertical** and **horizontal** scalability.
- First, you **upgrade the existing servers**, and later, **add more servers** when needed.

Example:

- Increase RAM first, and then add new servers as the business grows.

Used In:

- Large organizations or applications expecting continuous growth.

Diagram: 📌



In Short

Type	What It Does	Example
Vertical Scalability	Increases power of one machine	Add more RAM/CPU
Horizontal Scalability	Adds more machines	Add more servers
Diagonal Scalability	Mix of both	Upgrade + Add more servers



Summary

Cloud Scalability allows a system to **expand or reduce** resources easily.

It ensures **efficiency, cost savings, and smooth performance** as workloads change.

The three types — **Vertical, Horizontal, and Diagonal** — help meet different business needs.



Difference Between Cloud Scalability and Cloud Elasticity

Both **Scalability** and **Elasticity** are key features of cloud computing that help manage resources efficiently.

However, they differ in **how and when** resources are adjusted.



1. Basic Definition

- **Cloud Scalability:**

The ability to **increase or decrease resources** (like servers, storage, or power) to meet long-term needs.

- **Cloud Elasticity:**

The ability to **automatically adjust resources** up or down **instantly** based on real-time demand.



2. Nature

- **Scalability:** Planned and often **manual**.
- **Elasticity:** **Automatic and dynamic** — happens in real-time.



3. Purpose

- **Scalability:** To **handle gradual growth** of workloads or users over time.
- **Elasticity:** To **handle sudden changes** or spikes in workload.



4. Duration

- **Scalability:** Used for **long-term resource planning**.
- **Elasticity:** Used for **short-term demand fluctuations**.



5. Control

- **Scalability:** Controlled **manually** by administrators or engineers.
- **Elasticity:** Controlled **automatically** by the cloud system.



6. Cost Management

- **Scalability:** Helps manage cost over time by adding resources as the business grows.
- **Elasticity:** Saves cost by only using resources **when needed**.



7. Example Scenario

- **Scalability:** A company adds more servers as it gains more customers month by month.
- **Elasticity:** During a flash sale, the cloud automatically adds extra servers and removes them when traffic drops.



8. Implementation

- **Scalability:** Can be **vertical (up/down)** or **horizontal (out/in)**.
- **Elasticity:** Mostly involves **automatic scaling** based on system demand.



9. Reaction Time

- **Scalability:** Slower — changes made when growth is expected.
- **Elasticity:** Immediate — changes made the moment demand changes.



10. Used By

- **Scalability:** Businesses planning for **steady, predictable growth**.
- **Elasticity:** Businesses facing **variable or unpredictable workloads** (like e-commerce or streaming).



In Short (Summary Table)

Point	Cloud Scalability	Cloud Elasticity
1. Definition	Adjusts resources for long-term growth	Adjusts resources instantly based on demand
2. Nature	Manual or planned	Automatic and dynamic
3. Purpose	Handles steady growth	Handles sudden spikes
4. Duration	Long-term	Short-term

5. Control	Manual	Automatic
6. Cost	Grows with business	Pay only for what you use
7. Example	Adding more servers as users increase	Auto-adding servers during a flash sale
8. Type	Vertical & Horizontal scaling	Auto scaling
9. Reaction Time	Slow or planned	Instant or real-time
10. Best For	Predictable workloads	Unpredictable workloads

✓ Simple Summary

- **Scalability** = Planned growth → "Add more when needed."
- **Elasticity** = Instant change → "Add or remove automatically."

☁ What is On-Demand Provisioning in Cloud Computing?

On-demand provisioning means **getting computing resources instantly whenever you need them** — without manual setup or delay.

In simple words —

It allows users to **create, use, and manage cloud resources (like servers, storage, and software)** whenever they want, and **release them when they're no longer needed**.

⚙ How It Works

1. The user logs into a **cloud service provider** (like AWS, Azure, or Google Cloud).
2. They **choose the resources** they need (CPU, memory, storage, etc.).
3. The cloud system **automatically allocates** these resources within seconds.
4. When finished, the user can **stop or delete** the resources to save cost.

💡 Example

- Suppose you're launching a new website.
 - You need **extra servers** for testing.
 - You go to AWS, create virtual machines instantly, test your site, and delete them afterward.

👉 That's **on-demand provisioning** — fast, flexible, and cost-effective.

Types of On-Demand Provisioning

◆ 1. Server Provisioning

- Instantly create virtual servers (VMs) or containers.
- Example: Launching a new web server in seconds.

◆ 2. Storage Provisioning

- Allocate storage space only when needed.
- Example: Expanding your cloud storage for backup automatically.

◆ 3. Network Provisioning

- Configure virtual networks, firewalls, and load balancers instantly.
- Example: Setting up a secure network for your app.

Features of On-Demand Provisioning

- ⚡ **Instant Availability:** Get resources in seconds.
- 🔄 **Scalable:** Increase or decrease resources as needed.
- 💰 **Pay-as-You-Go:** Pay only for the resources you use.
- 🧠 **Automated Setup:** No manual installation or hardware setup.
- ⌚ **Time Saving:** Reduces the time to deploy applications.

Advantages of On-Demand Provisioning

- ✅ **Flexibility:** Users can quickly adjust to business needs.
- ✅ **Cost Efficiency:** No need to buy or maintain expensive hardware.
- ✅ **Speed:** Applications and services can be deployed faster.
- ✅ **Resource Optimization:** Use only what you need, when you need it.
- ✅ **Supports Innovation:** Developers can test and build new ideas easily.

Disadvantages

- ❌ **Overuse Risk:** If not monitored, costs can rise due to excessive usage.
- ❌ **Security Concerns:** Quick setup may lead to weak configurations if not managed properly.
- ❌ **Requires Internet:** You must be online to access resources.

In Short

On-Demand Provisioning means **getting cloud resources instantly whenever needed** — It helps users to **save time, reduce cost, and scale easily** using a **pay-as-you-go model**.



Example Cloud Services Offering On-Demand Provisioning

- Amazon Web Services (AWS EC2)
- Microsoft Azure Virtual Machines
- Google Cloud Compute Engine